Part A

Question 1:

$$D = \left\{ x \in \mathbb{R} : x = \frac{m}{p} \text{ for some } m \in \mathbb{N} \text{ and } p \text{ prime} \right\}$$

a)
$m \in \mathbb{N}$ is countable, Definition 6.4 HB Chapter 16 p87, and $p \in \mathbb{N}$ is countable. Hence $D \in \mathbb{Q}$, is countable, Theorem 6.6 HB Chapter 16 p87.

b)
i.
For

$$y \geq 0$$

and

$$r > 0$$

Hence

$$y + r > 0$$

Let

$$x = \frac{m}{p} \in D$$

such that

$$x > y$$

and

$$x < y + r$$

Then

$$y < \frac{m}{p} < y + r$$

$$yp < m < p(y + r)$$

$$\Rightarrow m \in (yp, p(y + r))$$

Since there are an infinite number of natural numbers, there exists at least one $m \in \mathbb{N}$ such that

$$m \in (yp, p(y + r))$$

Thus

$$y < \frac{m}{p} < y + r$$

and so

$$\exists x \in D : y < x < y + r$$

ii.

To show the D is dense in $[0, \infty)$, we need to show that for any

$$a, b \in [0, \infty)$$

such that

$$a < b$$

there exists

$$x \in D$$

such that

$$a < x < b$$

Let

$$r = b - a > 0$$

From part (i), we know that there exists

$$x \in D$$

such that

$$a < x < a + r$$

But

$$a + r = a + (b - a) = b$$

Thus

$$a < x < b$$

and so D is dense in $[0, \infty)$.